Unsupervised Contrastive Transformer Hashing for Cross-Modal Remote Sensing Retrieval

Zhan Yang, Weilu Long, Binghong Chen, Longzhi Sun, Yinan Li, and Xiaoyao Sun

***Abstract*—Deep-time Digital Earth (DDE) intends to harmonize the deep-time Earth data and open geoscience knowledge. To assist the objective, efficient cross-modal retrieval methodologies are essential to process heterogeneous Earth observation data. Hash learning maps diverse modalities of data into binary codes, providing a viable solution for large-scale remote sensing data retrieval, while the Transformer architecture enhances the accuracy and relevance of generated models by dynamically integrating knowledge in an unsupervised paradigm. In consequence, the combination of Transformer and hash learning replaces the dense vector search with Hamming distance, significantly reducing storage requirements and increasing retrieval efficiency via compact binary codes. In this article, we propose a novel unsupervised contrastive transformer hashing approach for cross-modal remote sensing retrieval (CTHRS). Initially, a Transformer-based multimodal fusion network is developed, which bridges the semantic gap between remote sensing images and texts via a self-attention mechanism and KL divergence alignment. Subsequently, a neighborhood structure-preserving mechanism is presented, which maintains the intramodal semantic coherence through dual similarity matrices derived from both original and hashing representations. Furthermore, a contrastive hash learning objective is designed, which combines intramodal decomposition and cross-modal alignment losses. Comprehensive experiments on RSICD, RSITMD, and UCM datasets demonstrate the superior performance of CTHRS state-of-the-art hashing approaches. The codes and datasets are available at https://github.com/holdibson/CTHRS**

***Index Terms*—Cross-modal retrieval, hashing learning, remote sensing, Transformer.**

NOMENCLATURE

$\mathbf{X}^v \in \mathbb{R}^{n\times d_v}$	Features of image modality.
$\mathbf{X}^t \in \mathbb{R}^{n\times d_t}$	Features of text modality.
$\mathcal{F}(\mathbf{X}, \theta)$	Multimodal fusion network.
$\mathcal{H}^v(\mathbf{X}^v, \theta^v)$	Image hashing network.
$\mathcal{H}^t(\mathbf{X}^t, \theta^t)$	Text hashing network.
$\mathbf{B} \in \{-1, 1\}^{n\times k}$	Hash code matrix.
n	Number of instances.
k	Length of hash code.

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Zhan Yang, Weilu Long, Binghong Chen, and Yinan Li are with the Big Data Institute, Central South University, Changsha 410083, China (e-mail: zyang22@csu.edu.cn; holdibson@csu.edu.cn; chenbinghong@csu.edu.cn; liyinan@csu.edu.cn).

Longzhi Sun is with the School of Computer Science and Engineering, Central South University, Changsha 410083, China (e-mail: sunlongzhi@csu.edu.cn).

Xiaoyao Sun is with the State Key Laboratory of Ocean Sensing, ZJU-Hangzhou Global Scientific and Technological Innovation Center, and the School of Earth Sciences, Zhejiang University, Hangzhou 310027, China (e-mail: sunxiaoyao@zju.edu.cn).

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I. INTRODUCTION

CROSS-MODAL remote sensing hashing (CMRSH) supports the quick location of specified remote sensing targets from textual descriptions or the retrieval of related metadata from a large number of unprocessed remote sensing images. With the iterative development of Earth observation technology, multimodal remote sensing data continue to increase, which exhibits significant spatiotemporal heterogeneity, such as surface mulching and target morphology from optical and SAR remote sensing images. Specifically, the remote sensing metadata possesses dynamic aggregation properties for large-scale storage, forming a multisource heterogeneous data system complementary to remote sensing images. Therefore, how to efficiently mine the potential value of diverse modalities of remote sensing data and achieve the transformation from data accumulation to knowledge discovery becomes the current focus of attention.

Kullback–Leibler divergence [1] is employed to characterize the similarity of feature distributions, realizing effective information interaction. Let $P(x)$ and $Q(x)$ be discrete probability distributions of variables. The information loss incurred when the expected distribution P is fit with the actual distribution Q can be formulated as follows:

$$D_{KL}(P \parallel Q) = \sum_x \left[P(x) \cdot \log\left(\frac{P(x)}{Q(x)}\right)\right]$$
$$s.t.\ D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$$

where the smaller the eigenvalue of $D_{KL}(P \parallel Q)$, the fewer the loss when Q approximates P.

Particularly, the BERT [2] is utilized in text feature extraction, which consists of discrete words, while remote sensing image features from ResNet [3] contain abundant semantic information. Note that the inherently convolutional inductive bias in the ResNet architecture generates locally coherent feature manifolds preserving translation invariance; meanwhile, the nonsparse activation patterns induced by the ReLU function and the regularization effects of BatchNorm layers jointly produce dense distributed representations. In contrast, the self-attention mechanism in BERT dynamically reweights the contribution of tokens, resulting in context-dependent feature

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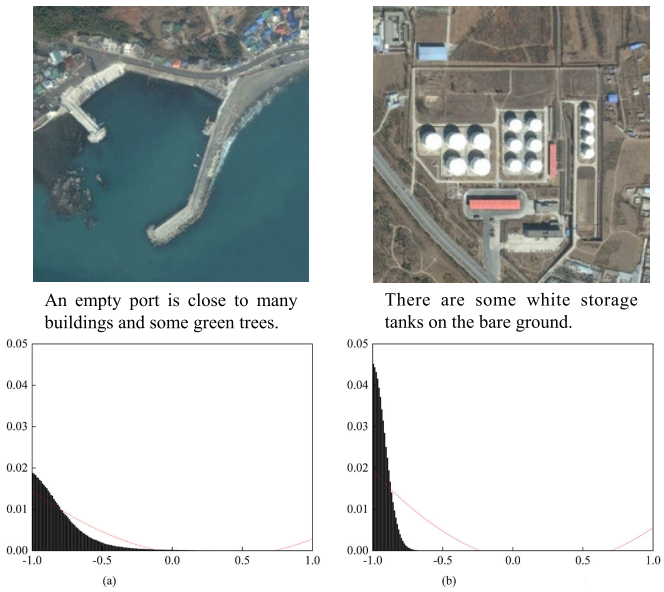


Fig. 1. Histogram distribution of (a) $\mathbf{S}_I$ and (b) $\mathbf{S}_T$ on the RSICD dataset.

distributions, which exhibit discrete semantic regions for polysemous words, with sparsity patterns emerging from attention heads that selectively focus on salient tokens. Moreover, the prevalent issues of text duplication and imprecision in remote sensing data amplify the comparative differences.

Let $\mathbf{S}_I$ and $\mathbf{S}_T$ denote the cosine similarity of remote sensing image and text features, respectively, and 2000 remote sensing image–text pairs are randomly selected from the RSICD [4] dataset. Furthermore, the statistical distributions of samples in the feature space are demonstrated in Fig. 1. It is obvious that $\mathbf{S}_I$ is dense, i.e., contains semantically relevant information, and $\mathbf{S}_T$ is sparse, where the left- and right-hand sides of $\mathbf{S}_I$ can be approximated as Gaussian distributions with significant differences. In order to improve the interactive capacity of the module, the remote sensing image feature is emphasized to filter out the noisy neighborhood correlations.

The rise of Transformer [5] architecture brings about a paradigm shift in cross-modal retrieval [6], [7], [8], [9], where the self-attention mechanism can effectively capture the long-distance dependence relations between heterogeneous modalities and exhibits special superiority in the semantic alignment of remote sensing data. Specifically, Transformer can not only model the complex mapping relations between spatial features of images and text semantics by constructing multilevel modal interaction networks but also dynamically adjust the weight distribution of diverse modal features, breaking through the limitation of manual feature assignment in conventional methods. In recent years, cross-modal pretrained models based on Transformer have performed noticeably in remote sensing scenes, such as object categorization and target description [10], [11], [12], demonstrating the superior representation capacity in dealing with high-dimensional heterogeneous data.

Cross-modal hashing [13], [14], [15], [16], [17], [18] is developing into a hot spot in large-scale remote sensing data mining, which can realize fast similarity retrieval based on binary codes through mapping heterogeneous data into a low-dimensional Hamming space, significantly reducing storage requirements and computation complexity. In order to implement the omnibearing description of objects, cross-modal retrieval of remote sensing data based on hash learning [19], [20], [21], [22] can instantly and accurately correlate diverse modalities in spatiotemporal dimensions, preserving the semantic similarity of heterogeneous modalities via Hamming distance, and possessing extensive application scenarios. However, the mainstream hashing approaches mainly focus on image retrieval [23], [24], [25], [26], [27] and encounter difficulty in the alignment accuracy and semantic gap of heterogeneous remote sensing features with complex modal discrepancies, which limit the further improvement of the retrieval paradigm. Particularly, the research on cross-modal image–text remote sensing hashing is scarce.

In consequence, it is essential to combine the efficient retrieval superiority of CMRSH with the in-depth semantic association capability of Transformer. On the one hand, the self-attention mechanism can explicitly adapt the fine-grained feature association of remote sensing modalities, providing accurate distribution alignment and semantic guidance for the learned hash codes. Concretely, utilizing the abundant visual characteristics to filter out the noise of text and enhance the cross-modal consistency, enabling the distribution of fusion features to approximate the semantic correlation of a real remote sensing scene. On the other hand, the binary encoding framework can compress the high-dimensional vectors generated from the Transformer into compact representations, satisfying the efficiency requirement for large-scale remote sensing retrieval. In other words, the fusion architecture achieves collaborative improvement of feature alignment, semantic preservation, and binary approximation through end-to-end joint optimization.

Existing CMRSH can be broadly separated into supervised and unsupervised approaches depending on whether semantic labels are used or not. Supervised methods leverage annotation information to reinforce the discriminative capacity of hash codes, which demonstrate superior retrieval performance. In contrast, unsupervised methods can only construct the training model through original data, resulting in diminished accuracy. However, most remote sensing data are unlabeled in practical applications, and it is time-consuming to annotate data via human analysts. Thus, utilizing the self-attention mechanism to explore the common latent representations of heterogeneous modalities and combining contrastive learning to construct a semantic-preserved hashing loss objective can effectively solve the issue of scarce labeled remote sensing data and provide a novel path for the intelligent retrieval of unsupervised cross-modal remote sensing data. Despite the satisfactory performance of existing unsupervised CMRSH methods, there exist subsequent key questions.

1) How to employ the abundant semantic information embedded in the visual modality to reinforce the sparse representation of text and extend to other modalities?
2) How to get the utmost of both intermodal and intramodal correlations through exploring the information

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interactions between diverse remote sensing modalities?

3) How to encode heterogeneous representations into the compact binary format while minimizing the semantic loss for cross-modal remote sensing image–text retrieval?

To maintain the information integrality between heterogeneous remote sensing modalities and perform efficient retrieval, we propose an unsupervised contrastive Transformer hashing approach for cross-modal remote sensing retrieval **(CTHRS)**. The contributions lie in threefold.

1) We propose a multimodal feature fusion network based on Transformer and alignment analysis of KL divergence, which enhances the semantic interaction between heterogeneous remote sensing modalities.
2) We present a neighborhood structure retention mechanism, ensuring unified semantic similarity in the hash codes' generation process.
3) We design a contrastive hashing loss to minimize the quantization error and further conduct comprehensive experiments to demonstrate the superior performance of **CTHRS** on cross-modal remote sensing retrieval tasks.

The remainder of this article is organized as follows. Initially, we survey prior works on CMRSH and then introduce the proposed **CTHRS** method in Section III. In Section IV, we contrast **CTHRS** with SOTA approaches and conduct ablation experiments to evaluate the core components of **CTHRS**. Finally, the conclusions are summarized in Section V.

II. Related Work

In this section, we review the typical hash learning and related cross-modal remote sensing retrieval approaches.

A. Cross-Modal Remote Sensing Image Hashing

CMRSH investigates the semantic relevance between heterogeneous remote sensing modalities in a common Hamming space, enabling accurate feature fusion and deep-level semantic sharing. In the earlier approaches, by learning nonlinear hash functions, both the hash codes of the database and the query are generated in a symmetric manner, resulting in significant time overhead due to the increased remote sensing data.

To further optimize the efficiency of remote sensing image retrieval, ADSH [28] proposes an asymmetric deep supervised hashing algorithm, where the hash function is learned only for the query set, while the hash codes of the database are directly generated. Subsequently, AHCL [29] presents an asymmetric hash learning strategy, which combines the semantic information of instances and similarity loss between remote sensing image pairs. Although asymmetric remote sensing image hashing algorithms possess superior properties in model training, there exists a lot of room for improvement in the retrieval accuracy. Furthermore, DHNNs [30] utilize a deep hashing architecture to perform preliminary exploration of large-scale remote sensing image retrieval. InterPDC [31] employs cross-category pairwise discrete and intraclass pairwise balanced cues to construct hash code books for sketches and remote sensing images, respectively. MsEspH [32] constructs a high-resolution SAR dataset and designs a deep hashing method with multisensor fusion and semantic preservation for image retrieval. DBHT [33] develops a distillation-based hashing Transformer approach, dealing with complex geometric representations in vessel images. In addition to image retrieval, CMRSH is used in audio-to-image retrieval tasks. DQH [34] proposes a quaternary hashing strategy to learn the semantic similarity relations between audio and remote sensing images. FADH [35] presents a fine-aligned discriminative hashing method for learning detailed information between remote sensing images and audio to generate high-quality hash codes.

With the advanced representation competency of deep neural networks, recent CMRSH methods can accurately preserve the semantic relevance between multimodal remote sensing data and perform superiorly in feature alignment. However, CMRSH pays insufficient attention to image and text modalities.

B. Cross-Modal Remote Sensing Image–Text Hashing

Text-based object retrieval intends to accurately search for specified targets from large-scale remote sensing images, and vice versa, which integrates the detection of candidate frames and association of visual–semantic information into a unified framework, enabling efficient end-to-end learning. CMRSH for image–text combines the neural network with hash learning to jointly optimize feature mapping and binary codes' generation, describing the instances from multiple perspectives.

By integrating original neighborhood information, DJSRH [36] captures the intrinsic semantic affinity between multimodal data and reconstructs the joint semantic relation via hash codes. JDSH [37] preserves the semantic relevance between heterogeneous modalities through constructing a joint-modal similarity matrix. Specifically, DGCPN [38] deals with graph neighbor consistency based on diverse modalities and neighborhood information. Furthermore, SAM [39] proposes a semantic alignment strategy to establish the potential relations between remote sensing image–text pairs and utilizes gates and attention mechanisms for nonnegligible feature optimization. GaLR [40] designs a multilevel dynamic fusion module, which employs local and global information to generate the discriminative binary codes for remote sensing text–image retrieval. Besides, DUCH [41] presents a contrastive hashing network, realizing effective similarity preservation in intra- and inter-remote sensing image–text. In addition, semi-paired data are a nonnegligible issue. CHNR [42] deals with noisy remote sensing image–text correspondences, utilizing a portion of available data to train the noise relation detection module via adversarial learning. DMFH [43] proposes a deep remote sensing hashing method, which extracts multiscale features from images and fine-grained semantic information from texts, respectively. Moreover, CACH [44] enhances the intramodal and intermodal semantic consistency via multiscale semantic similarity constraints and contrastive learning. Although the mentioned methods have achieved superior performance, the efficient scheme for minimizing the semantic gap between remote sensing text–image is the focus of cross-modal research.

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III. Method

A. Notations

For a clear illustration, this article mainly analyzes image and text modalities. Specifically, matrices and vectors are denoted by bold uppercase and lowercase letters, respectively, and the transpose of matrix $\mathbf{N}$ is denoted by $\mathbf{N}^\top$. Assume the multimodal database $\mathbf{X}^v$ and $\mathbf{X}^t$ with n remote sensing image–text pairs, where $\mathbf{x}_i^v$ and $\mathbf{x}_i^t$ are the ith sample of image and text, respectively, with feature dimensions d_v and d_t. The proposed method focuses on learning hash codes $\mathbf{B} \in \{-1, 1\}^{n\times k}$, where k is the length of hash code.

To generate compact binary representation, the multimodal fusion and hash networks for image and text modalities are formulated as $\mathcal{F}(\mathbf{X}, \theta)$, $\mathcal{H}^v(\mathbf{X}^v, \theta^v)$, and $\mathcal{H}^t(\mathbf{X}^t, \theta^t)$, where θ is the modal fusion network parameters to be learned, and θ^v and θ^t are the modality-specific hash network parameters. Moreover, the outputs of hash networks are defined as $\mathbf{H}^v = \mathcal{H}^v(\mathbf{X}^v, \theta^v)$ and $\mathbf{H}^t = \mathcal{H}^t(\mathbf{X}^t, \theta^t)$ for image and text modalities, where $\mathbf{h}_i^v$ and $\mathbf{h}_i^t$ are the ith sample, respectively. The main notations are listed in the Nomenclature.

B. Cross-Modal Similarity Matrix Construction

The remote sensing features extracted from deep neural networks after Transformer contain abundant semantic associations, which are of great importance to the CMRSH task with both image $\mathbf{F}^v \in \mathbb{R}^{n\times d_v}$ and text $\mathbf{F}^t \in \mathbb{R}^{n\times d_t}$ carrying complementary information. Particularly, the intramodal and intermodal similarity matrices under individual fusion-specific representations can be constructed via cosine similarity, which are utilized to characterize the neighborhood structure and cross-modal semantic relation of remote sensing image–text pairs, preserving the internal consistency

$$\begin{aligned}
\mathbf{S}_v^F &= \cos\left(\mathbf{F}_i^v, \mathbf{F}_j^v\right) = \frac{\mathbf{F}_i^v \mathbf{F}_j^{v\top}}{\| \mathbf{F}_i^v \|_2 \ \| \mathbf{F}_j^v \|_2} \\
\mathbf{S}_t^F &= \cos\left(\mathbf{F}_i^t, \mathbf{F}_j^t\right) = \frac{\mathbf{F}_i^t \mathbf{F}_j^{t\top}}{\| \mathbf{F}_i^t \|_2 \ \| \mathbf{F}_j^t \|_2} \\
\mathbf{S}^F &= \cos\left(\mathbf{F}_i^v, \mathbf{F}_j^t\right) = \frac{\mathbf{F}_i^v \mathbf{F}_j^{t\top}}{\| \mathbf{F}_i^v \|_2 \ \| \mathbf{F}_j^t \|_2}.
\end{aligned} \tag{1}$$

It is obvious that calculating the cosine similarity in a unified representation can explicitly evaluate whether the semantic alignment of remote sensing image–text embedding corresponds to each other, ensuring that similar instances remain in proximity within the respective feature spaces.

Moreover, compact hash codes that maintain the neighborhood structure of fusion representations can effectively improve the retrieval accuracy, and the similarity matrix of hash features reflects the intrinsic relation of binary codes in Hamming space. Particularly, the hash features can be obtained from individual fusion-specific representations through hash networks of a multilayer perceptron (MLP) structure, where the process of constructing similarity matrices involves the identical calculation as the fusion representation, ensuring that the neighborhood structure of original features is preserved in the hash space, which is essential for maintaining the semantic integrity of both image and text features in the remote sensing scene. Specifically, the respective intramodal and intermodal hash feature similarity matrices can be derived as follows:

$$\begin{aligned}
&\mathbf{S}_v^H = \cos\left(\mathbf{H}_i^v, \mathbf{H}_j^v\right), \quad \mathbf{S}_t^H = \cos\left(\mathbf{H}_i^t, \mathbf{H}_j^t\right) \\
&\mathbf{S}^H = \cos\left(\mathbf{H}_i^v, \mathbf{H}_j^t\right).
\end{aligned} \tag{2}$$

Notably, the intramodal hash similarity matrices are critical for preserving the structural information within each modality and bridging the semantic gap between visual and text features, ensuring that semantically similar features remain close in the hash space, which enables efficient matching of relevant remote sensing images with corresponding textual descriptions.

C. *CTHRS* Framework

As demonstrated in Fig. 2, the proposed **CTHRS** mainly contains three modules, i.e., multimodal semantic fusion based on Transformer (MSFT), intramodality decomposes contrastive loss (IDCL), and cross-modal hashing contrastive loss (CHCL). The details are given as follows.

1) MSFT: For remote sensing image–text pairs, *MSFT* extracts semantic information simultaneously, executes thorough feature interaction based on Transformer architecture, and further constructs the common cross-modal subspace. In view of inherent modal disparities, there exist distinct semantic distributions between remote sensing image and text features, which impact the priority and weight during modal fusion.

To facilitate efficient information fusion, it is essential to prealign the remote sensing image and text to filter out noisy neighborhood correlations. Consequently, we employ the abundant remote sensing image features to guide the sparse text, where the asymmetry is formally quantified before heterogeneous characteristics enter the Transformer fusion layer. Specifically, KL-divergence provides an asymmetric measure that penalizes missing modes in the target distribution, which is crucial for utilizing the pretrained representation of the source domain on the target domain such that

$$\min \mathcal{L}_{attn} = D_{KL}\left(P_{\mathbf{Z}} \parallel Q_{\mathbf{Z}}\right). \tag{3}$$

In addition, the combination of feature and distribution similarity through KL divergence can extract the common parts of heterogeneous modalities, enabling accurate semantic interaction. Let P^v be the probability distribution over cosine similarity for dense and high-variance remote sensing image features, P^t for sparse and low-variance text features. Since high-entropy distributions are easier to approximate than low-entropy ones such that broad P^v can cover narrow P^t without much penalty, P^t struggles to traverse P^v, which requires the probability mass in low-density regions.

On the other hand, P^v can be regarded as bimodal Gaussian mixture $P^v(x) = 0.5 \cdot \mathcal{N}(x; \mu = -0.5, \sigma_v^2) + 0.5 \cdot \mathcal{N}(x; \mu = 0.5, \sigma_v^2)$ and P^t as unimodal Gaussian centered at 0, where mass concentrated and more near zero due to duplication $P^t(x) = \mathcal{N}(x; \mu = 0, \sigma_t^2)$. Assume that centered Gaussians $P^t \sim \mathcal{N}(0, \sigma_t^2)$, and $P^v \sim \mathcal{N}(0, \sigma_v^2)$; with $\sigma_v > \sigma_t$, the

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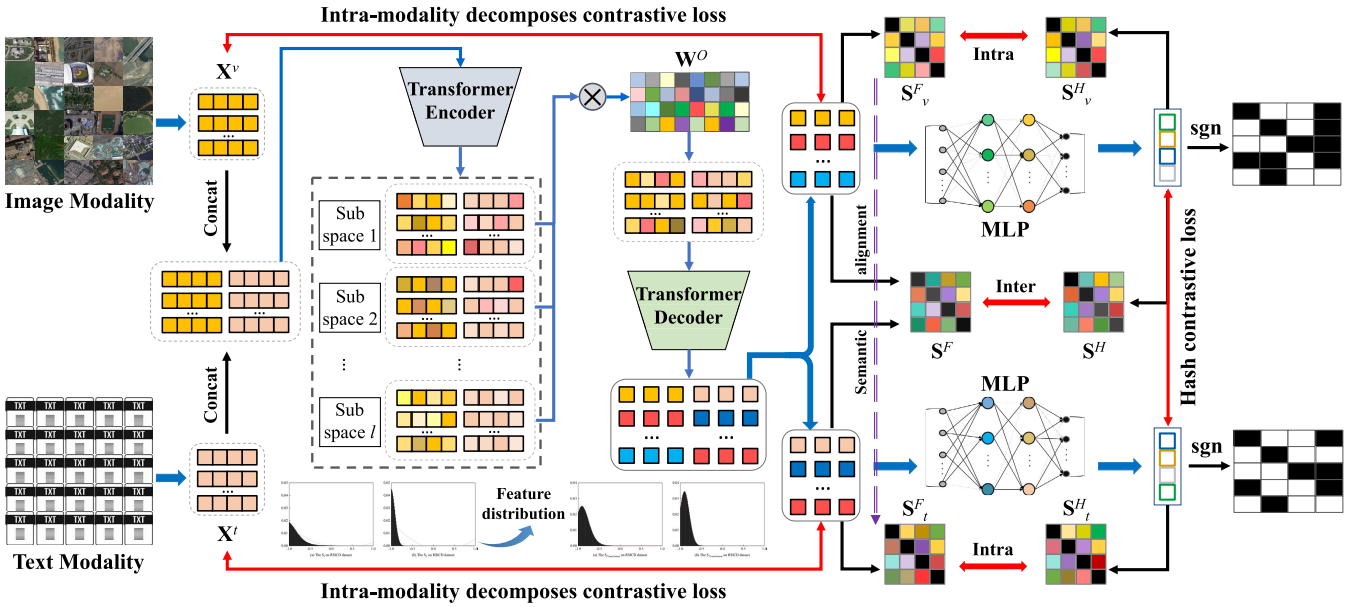


Fig. 2. Framework of the proposed unsupervised contrastive Transformer hashing for cross-modal remote sensing retrieval (**CTHRS**).

closed-form KL between $\mathcal{N}(\mu_{P^v}, \sigma^2_{P^v}) \parallel \mathcal{N}(\mu_{P^t}, \sigma^2_{P^t})$ can be formulated as follows:

$$D_{KL}\left(P^v \parallel P^t\right) = \log\left(\frac{\sigma_{P^t}}{\sigma_{P^v}}\right) + \frac{1}{2}\left[\frac{\sigma^2_{P^v} + (\mu_{P^v} - \mu_{P^t})^2}{\sigma^2_{P^t}} - 1\right] \tag{4}$$

where log term is positive and large when $\sigma_{P^t} > \sigma_{P^v}$, i.e., broad fitting narrow, negative when reversed. Similarly, the quadratic term $\gg 1$, dominates in the reverse. In other words, dense remote sensing images can filter out the noisy neighborhood correlation in sparse texts. Thus, we can deduce

$$D_{KL}\left(\mathbf{Z}^t \parallel \mathbf{Z}^v\right) < D_{KL}\left(\mathbf{Z}^v \parallel \mathbf{Z}^t\right). \tag{5}$$

Note that approximating dense and spread features with sparse and concentrated requires stretching P^t, incurring high penalties where P^v possesses mass, but P^t is near-zero density, which leads to poor alignment and information loss. In consequence, the remote sensing image-priority concatenation makes the self-attention mechanism propagate dense $\mathbf{Z}^v$ semantics to sparse $\mathbf{Z}^t$, minimizing loss in the low-KL direction.

Based on the above, assume that individual modality-specific features $\mathbf{X}^v$ and $\mathbf{X}^t$ can be approximately mapped to each other in the common space, and then, multimodal remote sensing features are concatenated into a unified representation

$$\begin{aligned} \mathbf{Z}^v &= \mathbf{X}^v\mathbf{W}^v + \mathbf{1}\mathbf{b}^v, \quad \mathbf{Z}^t = \mathbf{X}^t\mathbf{W}^t + \mathbf{1}\mathbf{b}^t \\ \mathbf{Z} &= \text{concat}\left[\mathbf{Z}^v; \mathbf{Z}^t\right] \in \mathbb{R}^{m\times(d_v+d_t)} \end{aligned} \tag{6}$$

where m is the batch size, $\mathbf{W}^v \in \mathbb{R}^{d_v\times\frac{d}{2}}, \mathbf{W}^t \in \mathbb{R}^{d_t\times\frac{d}{2}}$, $\mathbf{b} \in \mathbb{R}^{1\times\frac{d}{2}}$ denotes bias terms, and the adjustable variate controls the distribution weight of similarity judgment, facilitating the semantic association between heterogeneous modalities.

Furthermore, feeding the concatenation representation into the semantic fusion Transformer architecture, which employs the self-attention mechanism to capture intramodal dependencies and semantic correlations as follows:

$$\begin{aligned} \text{attention}\left(\mathbf{Q},\mathbf{K},\mathbf{V}\right) &= \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V} \\ \text{MutiHead}\left(\mathbf{Q},\mathbf{K},\mathbf{V}\right) &= \text{Concat}\left(head_1,\ldots,head_l\right)\mathbf{W}^O \\ s.t.\ head_i &= \text{attention}\left(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V\right) \end{aligned} \tag{7}$$

where $head_i$ is the ith attention head and $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ denote the query, key, and value matrices, respectively. Specifically, parallel attention heads $l = 4$ with the dimension of $d = d_v + d_t$ and learnable weight matrices $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V$ are head-specific parameters, which project the multimodal remote sensing semantic information into respective query, key, and value representations. Besides, $\mathbf{W}^O$ is a common parameter matrix that combines the concatenated outputs from attention heads to generate the ultimate transformed representation.

Particularly, the modality concatenation order influences how the self-attention captures dependencies. The abundant remote sensing image features as the primary reference token in the sequence, enables attention to propagate visual semantics to the sparse text feature, reduces information loss and noisy neighborhood correlations between heterogeneous modalities, is superior for semantic alignment and consistent postfusion distributions, and meets the requirements of remote sensing scenes.

In consequence, the *MSFT* module positions $\mathbf{Z}^v$ as the attention anchor, structurally guiding the model to prioritize the attention query process toward semantic correlations established by remote sensing image features. Furthermore, the sparse $\mathbf{Z}^t$ is required to align the key and value with the stable query space image-defined. This mechanism realizes the noise-filtering objective by ensuring that high-information visual context directs the semantic flow, which prevents sparse and unstable text features from dominating the global attention process and avoids attention being dispersed into discrete

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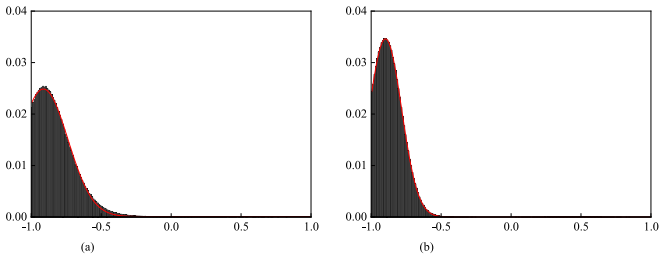


Fig. 3. Histogram distribution of (a) $\mathbf{S}_{I\text{Transformer}}$ and (b) $\mathbf{S}_{T\text{Transformer}}$ on the RSICD dataset after the *MSFT* module.

or imprecise text semantic regions. Specifically, the reverse concatenation order, while dimensionally feasible, is catastrophic for the modality fusion strategy. In view of the inherent sparsity and context dependence, the $\mathbf{Z}^t$ driven initial $\mathbf{Q}$ vectors would be highly unstable or diffuse.

As depicted in Fig. 3, the feature distributions of $\mathbf{S}_I$ and $\mathbf{S}_T$ on the RSICD dataset after MSFT become consistent in both remote sensing image and text than the original features in Fig. 1, realizing efficient information fusion and proving the validity of the *MSFT* module.

2) IDCL: After Transformer encoding, the unified remote sensing representation $\mathbf{Z}$ is processed through the modal fusion network to obtain the cross-modal semantic representation $\mathbf{F} = \mathcal{F}(\mathbf{Z}, \theta)$, which can be further decomposed into individual modality-specific representations $\mathbf{F}^v$ and $\mathbf{F}^t$, where $\mathbf{f}_i^v$ and $\mathbf{f}_i^t$ are the respective ith sample.

To guarantee that the fusion representation of modality-specific contains consistent semantics, the intramodality contrastive losses are, respectively, defined as follows:

$$\mathcal{L}_{fusion}^{v}\left(\mathbf{x}_i^v, \mathbf{f}_i^v\right) = -\sum_{i=1}^{m} \log \frac{\exp\left(\cos\left(\mathbf{x}_i^v, \mathbf{f}_i^{v+}\right)/\tau\right)}{\sum_{k=1}^{m} \exp\left(\cos\left(\mathbf{x}_i^v, \mathbf{f}_k^v\right)/\tau\right)}$$
$$\mathcal{L}_{fusion}^{t}\left(\mathbf{x}_i^t, \mathbf{f}_i^t\right) = -\sum_{i=1}^{m} \log \frac{\exp\left(\cos\left(\mathbf{x}_i^t, \mathbf{f}_i^{t+}\right)/\tau\right)}{\sum_{k=1}^{m} \exp\left(\cos\left(\mathbf{x}_i^t, \mathbf{f}_k^t\right)/\tau\right)} \tag{8}$$

where τ denotes the temperature coefficient and $\mathbf{x}_i^v, \mathbf{f}_i^{v+}$ and $\mathbf{x}_i^t, \mathbf{f}_i^{t+}$ denote truly aligned remote sensing image–text pairs. Accordingly, the ultimate modal fusion contrastive loss $\mathcal{L}_{C_{inter}}$ is defined as follows:

$$\mathcal{L}_{C_{fusion}} = \mathcal{L}_{fusion}^{v} + \mathcal{L}_{fusion}^{t}. \tag{9}$$

In general, the semantic information associated with hash codes deviates from the remote sensing representations, while the hash similarity matrix derived from the joint-modal similarity matrix contains the original neighborhood structure. Moreover, with regard to modal semantic alignment, $\mathbf{S}^F$ and $\mathbf{S}^H$ serve as the cross-modal similarity matrix of fusion representation and hash feature, respectively, ensuring that similarity relations between remote sensing image–text pairs in the Hamming space are consistent with those in the original feature space. Therefore, the loss of similarity matrices is designed as follows:

$$\mathcal{L}_Q = \frac{1}{2}\left(\left|\left|\mathbf{S}_v^F - \mathbf{S}_v^H\right|\right|^2 + \left|\left|\mathbf{S}_t^F - \mathbf{S}_t^H\right|\right|^2\right) + \frac{1}{2}\left(\left|\left|\eta\mathbf{S}^F - \mathbf{S}_v^H\right|\right|^2 + \left|\left|\eta\mathbf{S}^F - \mathbf{S}_t^H\right|\right|^2\right) \tag{10}$$

$$\mathcal{L}_I = \left|\left|\eta\mathbf{S}^F - \mathbf{S}^H\right|\right|^2 + \sum_{i=1}^{m}\left|\left|\eta \cdot 1 - \text{diag}\left(s_{ii}^H\right)\right|\right|^2 \tag{11}$$

where $\text{diag}(s_{ii}^H)$ is the diagonal element of $\mathbf{S}^H$ and $\eta = 1.5$ is a scaling factor determined on the observations: 1) $\eta = 1$, the model degenerates to a trivial solution where all samples share an identical hash code and 2) $\eta = 2$, hash codes become overly dispersed, disrupting the inherent semantic structure.

As the self-similarity regularization, which forces the hash code similarity of each sample to be η, ensuring that the model is unable to learn proprietary hash codes as the same vector, thereby losing discriminative ability. Particularly, inspired by spectral hashing, the loss is devised to preserve the neighborhood structure embedded in the original remote sensing features while preventing the emergence of trivial solutions.

Although the Transformer architecture facilitates feature interaction, the intrinsic distributional heterogeneity between modalities remains a challenge. Remote sensing images possess dense and high-variance semantic distributions, whereas text descriptions typically exhibit sparse patterns due to the finite vocabulary. To explicitly bridge the "distribution gap" and enforce semantic consistency before hashing, we present a distribution-consistent alignment mechanism using KL divergence.

Specifically, after obtaining the fusion representations $\mathbf{F}^v$ and $\mathbf{F}^t$, the intramodal similarity matrices $\mathbf{S}_v^F$ and $\mathbf{S}_t^F$ are constructed to represent the neighborhood structure within each modality. To make the distributions comparable, we convert the similarity scores into probability distributions via a softmax function as follows:

$$s_{ij}^v = \frac{\exp\left(\mathbf{S}_{v,ij}^F/\tau\right)}{\sum_k \exp\left(\mathbf{S}_{v,ik}^F/\tau\right)}, \quad s_{ij}^t = \frac{\exp\left(\mathbf{S}_{t,ij}^F/\tau\right)}{\sum_k \exp\left(\mathbf{S}_{t,ik}^F/\tau\right)} \tag{12}$$

where s_{ij}^v and s_{ij}^t represent the probability of the jth instance being a semantic neighbor of the ith instance in the remote sensing image and text manifolds, respectively.

Leveraging the observation that visual features contain abundant semantic information, we treat the image distribution as the "soft target" to guide the optimization of the text distribution. Thus, we minimize the KL divergence $D_{KL}(\mathbf{s}_i^v \parallel \mathbf{s}_i^t)$ to force the text representation to capture the comprehensive semantic topology inherent in the images. The regularization loss can be formulated as follows:

$$\mathcal{L}_{KL} = \sum_{i=1}^{m} D_{KL}\left(\mathbf{s}_i^v \parallel \mathbf{s}_i^t\right) = \sum_{i=1}^{m}\sum_{j=1}^{m} s_{ij}^v \log\left(\frac{s_{ij}^v}{s_{ij}^t}\right). \tag{13}$$

By minimizing $\mathcal{L}_{KL}$, we compel the sparse text modality to expand its semantic coverage, approximating the dense neighborhood relations of visual data, which ensures that the fused representations maintain structural consistency between modalities, preventing the "collapse" of text features into isolated clusters and generating more robust hash codes.

3) CHCL: For the quantization features of remote sensing image–text, the intramodal and intermodal hash similarity matrices can characterize the neighborhood correlation

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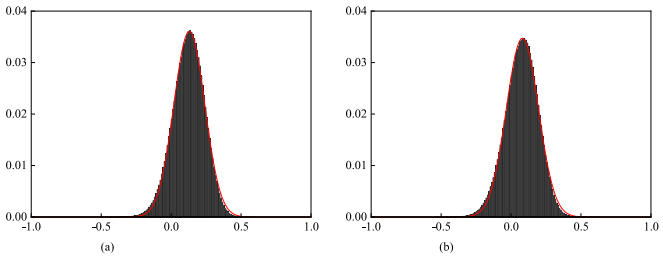


Fig. 4. Histogram distribution of $\mathbf{S}_I$ and $\mathbf{S}_T$ on the RSICD dataset after *CHCL* module. (a) H_I and (b) H_T on the RSICD dataset.

Algorithm 1 Proposed **CTHRS**

Require: Remote sensing training set $\mathbf{X}^v$ and $\mathbf{X}^t$, batch size m, maximum epoch number ω, hyper-parameters $\alpha, \beta, \lambda, \gamma, \eta$.

Ensure: Neural network parameters of the to-be-learned multimodal semantic fusion network θ, and hash networks θ^v, θ^t.

Initialize the parameters θ, θ^v, θ^t.
For $i \in \left[1, \frac{n}{m}\right]$ **do**
  Randomly select m training remote sensing image-text pairs.
  Generate $\mathbf{F}^v$ and $\mathbf{F}^t$ via forward propagation.
  Process $\mathbf{F}^v$ and $\mathbf{F}^t$ to obtain $\mathbf{S}_v^F, \mathbf{S}_t^F, \mathbf{S}^F$ via Eq. (1).
  Calculate intra-modality decomposes contrastive loss via Eq. (9).
  Generate $\mathbf{H}^v$ and $\mathbf{H}^t$ via forward propagation.
  Process $\mathbf{H}^v$ and $\mathbf{H}^t$ to obtain $\mathbf{S}_v^H, \mathbf{S}_t^H, \mathbf{S}^H$ via Eq. (2).
  Calculate similarity matrix loss via Eqs. (10), (11).
  Calculate KL divergence loss of distributions via Eq. (13).
  Calculate cross-modal hashing contrastive loss via Eq. (14).
  Calculate the objective function Eq. (15), and update parameters θ, θ^v, θ^t via back propagation.
Until up to ω.
End For
Return θ, θ^v, θ^t.

between the hash features of heterogeneous modalities. Thus, the loss can be calculated as follows:

$$\mathcal{L}_{C_{inter}}\left(\mathbf{h}_i^v, \mathbf{h}_i^t\right) = -\sum_{i=1}^{m} \log \frac{\exp\left(\cos\left(\mathbf{h}_i^v, \mathbf{h}_i^{t+}\right)/\tau\right)}{\sum_{k=1}^{m} \exp\left(\cos\left(\mathbf{h}_i^v, \mathbf{h}_k^t\right)/\tau\right)} \quad (14)$$

where $\mathbf{h}_i^v$ and $\mathbf{h}_i^t$ denote the hash features from hash network and $\mathbf{h}_i^v, \mathbf{h}_i^{t+}$ is truly aligned remote sensing image–text pair. Furthermore, $\mathbf{B} = \mathrm{sgn}(\mathbf{H})$ is suited for each modality, where $\mathrm{sgn}(\cdot)$ is the elementwise indicator operator.

As depicted in Fig. 4, the feature distribution of the RSICD dataset after CHCL becomes particularly consistent in both remote sensing image and text, which proves the validity of the hashing loss module.

Notably, to generate the hash code for a query, its original feature is processed through the modality-specific part of the trained network. Then, the retrieval is performed by calculating the Hamming distance between the query hash code and all the hash codes in the database. In other words, the proposed **CTHRS** learns to harmonize heterogeneous modalities using paired remote sensing data, which emphasizes the ability to generalize semantic alignment to foreign inputs, enabling valid performance evaluation.

In consequence, the unsupervised training of deep hash networks can be significantly improved through guiding the generation of hashing similarity matrices via associated modal similarity matrices, which learns the neighborhood structure of remote sensing data such that the generated hash codes possess a high level of original correlation in terms of instances, further maintaining as much semantic information as possible.

D. Overall Loss Function

The similarity matrix involves the semantic alignment of intramodal and intermodal fusion representations and hashing features, where the neighborhood structure of original features is learned through minimizing the reconstruction error between similarity matrices, and the intrinsic semantic correlations between heterogeneous remote sensing modalities are explored to compensate for the lack of semantic information after conversion to hash features. In consequence, integrating (9)–(11), (13), and (14) into a unified learning framework, the overall loss function for hash codes learning can be derived as follows:

$$\min_{\theta,\theta^v,\theta^t} \mathcal{L} = \alpha \mathcal{L}_{C_{fusion}} + \beta\left(\mathcal{L}_Q + \mathcal{L}_I\right) + \lambda \mathcal{L}_{KL} + \gamma \mathcal{L}_{C_{inter}}. \quad (15)$$

Particularly, we summarize the comprehensive procedure of **CTHRS** in Algorithm 1.

IV. Experiment

A. Experiment Settings

1) Datasets: In this article, we employ RSICD, RSITMD [45], and UCM [46] datasets for evaluation, where images and texts use 512-D ResNet and 512-D BERT features, respectively. Particularly, each remote sensing image is sized at 224 $\times$ 224 pixels and possesses five relevant semantic captions. For the **RSICD** dataset, we randomly choose 5460/1092 image–text pairs as the training/query sets from 31 categories, and the residuals are utilized as the retrieval set. For the **RSITMD** dataset, we randomly select 2371/474 image–text pairs as the training/query sets, while the residuals with 32 tags are used as the retrieval set. Similarly, the **UCM** dataset is randomly divided into 1050/210 training/query sets with 21 labels, and the residual image–text pairs are used as the retrieval set.

2) Baselines and Evaluation Metrics: In the experiments, unsupervised cross-modal remote sensing hashing methods, DUCH [41], CHNR [42], and DMFH [43], and routine hashing approaches, JDSH [37], DGCPN [38], CIRH [47], UCCH [48], UCMFH [49], PT-FUCH [50], SMSH [51], and VTM-UCH [52], are compared with **CTHRS** in retrieving textual data by visual query (I2T) and retrieving visual data by text query (T2I).

In order to assess the retrieval performance of **CTHRS**, mean average precision (**mAP**), precision–recall (**PR**) curves, and top-k precision (**P**@k) act as evaluation metrics.

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3) Implementation Details: To maintain the consistency of baselines, we utilize a pretrained ResNet18 to extract remote sensing image features, further obtain embeddings with 512-D by removing the final fully connected layer. Besides, we sum the last four hidden states for each token in the sentence to derive embeddings with 768-D via pretrained BERT-base-uncased and further pass through a fully connected layer to obtain text features with 512-D. Specifically, the image net is designed as $d_v \rightarrow 4096 \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow k \rightarrow \tanh(\cdot)$, where ReLU and $\tanh(\cdot)$ are activation functions. The structure of the text net is consistent with the image net.

Particularly, the hyperparameters remain uniform on the three datasets during training, with the settings of $\alpha = 1.0$, $\beta = 0.1$, $\lambda = 0.01$, and $\gamma = 0.1$. We adopt an Adam optimizer with parameters betas $= (0.5, 0.999)$ for the three networks and set the learning rate of the fusion network to 0.0001. For image and text networks, we set the learning rates to 0.001, 0.001, and 0.002 on **RSICD**, **RSITMD**, and **UCM** datasets, respectively. Moreover, the batch size is fixed to 256, the total iteration number is set to 100, and the number of retrieval instances is set to 20. For the baselines, we realized their works via open-source codes. All experiments are trialed on a server with Intel Xeon Silver 4214 Processor @ 2.40 GHz, NVIDIA RTX 3080Ti GPU, 128-GB RAM.

B. Results

Table I manifests the mAP scores of each method on the three datasets with hash code lengths from 16 to 128 bits. The PR and top-k precision curves are demonstrated in Figs. 5 and 6, respectively. From Table I and Figs. 5 and 6, we can obtain the following key findings.

1) **CTHRS** surpassed all the baselines, especially on the short hash bits. For the I2T task, contrasted with the best baselines, the average mAP scores of **CTHRS** are beyond by 4.51%, 9.59%, and 5.42% on RSICD, RSITMD, and UCM datasets, respectively. There exist at least 10.83%, 10.11%, and 3.07% gains for the T2I task on the three datasets, respectively.
2) Although DUCH and DMFH are efficient cross-modal remote sensing hashing methods, **CTHRS** achieves superior retrieval performance, particularly on the T2I tasks. The improvement in mAP scores is attributed to employing the dual similarity matrices derived from both fusion-specific and hash representations, preserving the cross-modal semantic coherence and neighborhood structure of remote sensing data, which reinforces the discriminative competency of the model.
3) RSICD and RSITMD datasets involve high-resolution satellite images with complex scenes, such as land cover and multiscale objects paired with lengthy descriptions, where texts fail to capture visual details, which creates a larger semantic gap than general datasets, leading to noisy and incomplete alignments for routine methods.
4) By comparing SOTA approaches, VTM-UCH relies on CLIP for vision–text correspondence and K-means clustering for similarity matrices, but remote sensing images possess context-dependent features, such as viewpoint shifts and cloud cover, resulting in error-prone object detection.
5) The mAP scores of most methods increase with the increase in code lengths. In addition, the PR and top-k curves depict a similar trend to mAP scores with the increase in retrieved instances.

C. Ablation Experiments

In this section, we conduct ablation studies to validate the efficacy of core components within the proposed **CTHRS**.

1) Impacts of Cross-Modal Similarity: To verify the influence of cross-modal similarity matrices, variant **CTHRS**-$\mathcal{S}$ removed the intermodal similarity loss, i.e., (10) and (11), which exhibits performance degradation in Table II. The variant manifests that the cross-modal similarity matrix plays a nonnegligible role in preserving structural information within the learned hash codes, which reflects the intrinsic relation among binary codes in Hamming space, guaranteeing that similar image–text pairs maintain close proximity, further strengthening the semantic coherence between heterogeneous remote sensing modalities.

2) Impacts of IDCL: To validate the decisive effect of the multimodal semantic fusion component, variant **CTHRS**-$\mathcal{I}$ is presented, which removes the IDCL in (9). From Table II, there exists an obvious performance decrease in **CTHRS**-$\mathcal{I}$ on the three datasets, which disrupts the feature decoupling mechanism, driving the model to degenerate into an initial hashing network whose retrieval performance approximates random guessing. The variant indicates that cross-modal semantic interaction plays a critical guiding role in the hash learning process, which can effectively preserve the intramodal semantic information in binary codes.

3) Impacts of CHCL: To verify the impact of contrastive learning, **CTHRS**-$\mathcal{H}$ removed the CHCL in (15). Experiment results in Table II demonstrate that the proposed hash learning loss can effectively enhance the quality of generated hash codes, improving retrieval performance through maintaining multimodal neighborhood correlations within the hash space.

4) Impacts of Modality Concatenation Order: To validate the influence of modality concatenation order before heterogeneous characteristics entering the Transformer fusion layer, **CTHRS**-$\mathcal{C}$ exchanged the sequence direction such that

$$\mathbf{Z} = \text{concat}\left[\mathbf{Z}^t; \mathbf{Z}^v\right] \in \mathbb{R}^{m \times (d_t + d_v)}.$$

The reverse order prioritizes sparse texts, attempting to approximate the dense image distribution, which leads to higher loss and suboptimal information interaction, resulting in the dense semantics of remote sensing image features being diluted, and the alignment accuracy is decreased, further degrading the retrieval performance in cross-modal tasks within Table II.

D. Impacts of Vision-Language Model Features

Existing cross-modal hashing methods typically integrate deep neural networks to extract high-level features from multimodal data within end-to-end optimization. Datasets are

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TABLE I

PERFORMANCE COMPARISON (MAP) OF **CTHRS** AND BASELINES ON THE THREE DATASETS WITH VARIOUS CODE LENGTHS

Task	Method	RSICD				RSITMD				UCM			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
I → T	DUCH	0.684	0.791	0.836	0.829	0.306	0.420	0.515	0.577	0.760	0.794	0.844	0.870
	CHNR	0.583	0.703	0.730	0.781	0.466	0.609	0.708	0.740	0.707	0.764	0.842	0.850
	DMFH	0.703	0.746	0.841	0.849	0.673	0.719	0.761	0.826	–	–	–	–
	JDSH	0.627	0.722	0.772	0.810	0.519	0.644	0.719	0.765	0.707	0.763	0.840	0.862
	DGCPN	0.511	0.703	0.776	0.792	0.414	0.501	0.562	0.599	0.308	0.517	0.609	0.625
	CIRH	0.639	0.678	0.702	0.713	0.577	0.649	0.665	0.670	0.708	0.717	0.770	0.757
	UCCH	0.597	0.722	0.815	0.844	0.468	0.683	0.784	0.820	0.663	0.793	0.877	0.889
	UCMFH	0.523	0.637	0.664	0.636	0.387	0.464	0.500	0.483	0.480	0.599	0.683	0.727
	PT-FUCH	0.542	0.666	0.720	0.799	0.434	0.447	0.647	0.725	0.712	0.763	0.834	0.869
	SMSH	0.600	0.622	0.634	0.684	0.556	0.613	0.677	0.652	0.694	0.822	0.832	0.825
	VTM-UCH	0.505	0.607	0.572	0.604	0.488	0.524	0.518	0.555	0.567	0.658	0.661	0.659
	CTHRS	**0.778**	**0.841**	**0.847**	**0.852**	**0.777**	**0.823**	**0.835**	**0.842**	**0.851**	**0.873**	**0.895**	**0.902**
T → I	DUCH	0.697	0.780	0.824	0.826	0.288	0.350	0.444	0.516	0.799	0.851	0.916	0.927
	CHNR	0.601	0.746	0.783	0.836	0.454	0.597	0.713	0.740	0.729	0.840	0.904	0.917
	DMFH	0.712	0.793	0.844	0.852	0.692	0.734	0.761	0.792	–	–	–	–
	JDSH	0.694	0.787	0.846	0.888	0.558	0.672	0.744	0.789	0.684	0.810	0.841	0.865
	DGCPN	0.506	0.747	0.837	0.860	0.393	0.535	0.607	0.647	0.324	0.500	0.619	0.659
	CIRH	0.640	0.714	0.732	0.752	0.571	0.662	0.694	0.702	0.701	0.745	0.811	0.777
	UCCH	0.636	0.757	0.845	0.867	0.522	0.727	0.826	0.854	0.705	0.808	0.883	0.879
	UCMFH	0.523	0.618	0.643	0.615	0.389	0.455	0.490	0.453	0.504	0.654	0.682	0.729
	PT-FUCH	0.594	0.762	0.857	0.886	0.421	0.626	0.718	0.761	0.720	0.781	0.845	0.870
	SMSH	0.572	0.601	0.630	0.719	0.488	0.620	0.645	0.748	0.686	0.802	0.848	0.824
	VTM-UCH	0.517	0.650	0.577	0.610	0.496	0.539	0.522	0.535	0.624	0.672	0.662	0.624
	CTHRS	**0.834**	**0.912**	**0.917**	**0.925**	**0.801**	**0.857**	**0.869**	**0.877**	**0.858**	**0.887**	**0.918**	**0.931**

TABLE II

MAP RESULTS OF **CTHRS** AND VARIANTS ON RSICD, RSITMD, AND UCM DATASETS

Task	Method	RSICD				RSITMD				UCM			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
I → T	**CTHRS**-$\mathcal{S}$	0.769	0.823	0.825	0.821	0.759	0.818	0.825	0.826	0.840	0.868	0.883	0.891
	CTHRS-$\mathcal{I}$	0.062	0.062	0.062	0.062	0.059	0.059	0.059	0.059	0.132	0.132	0.132	0.132
	CTHRS-$\mathcal{H}$	0.769	0.833	0.836	0.840	0.723	0.783	0.797	0.792	0.830	0.866	0.881	0.870
	CTHRS-$\mathcal{C}$	0.767	0.834	0.832	0.842	0.727	0.786	0.823	0.817	0.793	0.838	0.853	0.888
	CTHRS	**0.778**	**0.841**	**0.847**	**0.852**	**0.777**	**0.823**	**0.835**	**0.842**	**0.851**	**0.873**	**0.895**	**0.902**
T → I	**CTHRS**-$\mathcal{S}$	0.823	0.902	0.905	0.912	0.792	0.851	0.864	0.872	0.815	0.871	0.881	0.893
	CTHRS-$\mathcal{I}$	0.062	0.062	0.062	0.062	0.059	0.059	0.059	0.059	0.132	0.132	0.132	0.132
	CTHRS-$\mathcal{H}$	0.814	0.907	0.899	0.912	0.757	0.802	0.827	0.836	0.835	0.882	0.909	0.920
	CTHRS-$\mathcal{C}$	0.804	0.899	0.908	0.910	0.763	0.835	0.860	0.849	0.804	0.864	0.894	0.891
	CTHRS	**0.834**	**0.912**	**0.917**	**0.925**	**0.801**	**0.857**	**0.869**	**0.877**	**0.858**	**0.887**	**0.918**	**0.931**

reinforced with pretrained vision-language models, emphasizing scalability and generalization in high-dimensional spaces. Thus, it is essential to analyze how vision-language model features influence retrieval performance. To ensure the fairness of experimental settings, we employ the CLIP feature extracted from 400 million image–text pairs [53] for training and testing on the RSICD dataset, termed **CTHRS-V**.

Baselines within the CLIP feature are evaluated in comparison with **CTHRS-V**. The mAP results are detailed in Table III, revealing that **CTHRS-V** achieves superior mAP scores compared to baselines. The observation of DGCPN, CIRH, and UCCH underscores the contribution of a pretrained vision-language model in enhancing the visual–semantic representation within the generated hash codes. However, the CLIP-based mAP scores of DUCH and CHNR have declined, demonstrating that direct application of general VLMs requires adaptation or fine-tuning for remote sensing tasks to avoid domain shift. Besides, SMSH is designed to analyze distinct features between heterogeneous modalities via statistical law, whereas the CLIP features for remote sensing images and text are similar. Thus, it cannot learn valuable information.

E. Efficiency Analysis

This section evaluates the efficiency of the proposed method compared to baselines on the RSICD dataset. Table IV provides training/inference times (s) and memory (MB) in 100 epochs with a code length of 64 bits. It is observed that **CTHRS** needs more training time than conventional approaches, which is a common phenomenon due to the multilayer Transformer iteration optimization. In contrast, the inference time is less than that of most methods. Furthermore, compared to the DNN-based DGCPN, we need less time in the training stage, and the memory is in an acceptable range.

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TABLE III
MAP RESULTS OF **CTHRS-V** AND BASELINES WITHIN CLIP FEATURE ON THE RSICD DATASET

Task	Method	RSICD			
		16 bits	32 bits	64 bits	128 bits
I → T	DUCH	0.626	0.736	0.791	0.788
	CHNR	0.750	0.788	0.778	0.816
	JDSH	0.310	0.408	0.506	0.604
	DGCPN	0.688	0.828	0.891	0.901
	CIRH	0.739	0.818	0.850	0.866
	UCCH	0.785	0.894	0.929	0.936
	UCMFH	0.802	0.854	0.893	0.899
	PT-FUCH	0.665	0.695	0.705	0.706
	SMSH	0.059	0.070	0.093	0.048
	VTM-UCH	0.532	0.665	0.671	0.705
	CTHRS-V	**0.923**	**0.935**	**0.939**	**0.943**
T → I	DUCH	0.663	0.758	0.801	0.806
	CHNR	0.767	0.795	0.804	0.837
	JDSH	0.356	0.394	0.488	0.557
	DGCPN	0.717	0.777	0.858	0.862
	CIRH	0.749	0.785	0.800	0.802
	UCCH	0.757	0.851	0.889	0.891
	UCMFH	0.779	0.858	0.890	0.879
	PT-FUCH	0.666	0.699	0.726	0.714
	SMSH	0.058	0.065	0.079	0.078
	VTM-UCH	0.547	0.694	0.715	0.738
	CTHRS-V	**0.897**	**0.908**	**0.914**	**0.919**

TABLE IV
EFFICIENCY COMPARISON OF **CTHRS** AND BASELINES ON RSICD

Method	RSICD		
	Training time	Inference time	Memory
DUCH	75.26	0.14	499
CHNR	44.82	0.12	557
JDSH	103.84	0.28	617
DGCPN	272.57	0.26	1,876
CIRH	40.13	0.11	1,823
UCCH	62.88	0.44	2,081
UCMFH	98.83	0.20	1,483
PT-FUCH	8050.50	1.39	3,941
SMSH	125.81	0.38	1,711
VTM-UCH	63.53	0.10	3,173
CTHRS	**180.54**	**0.14**	**4,595**

The slight added complexity in the fusion module yields significant accuracy gains, with **CTHRS** outperforming baselines by 3.73%–12.98% in mAP on the three datasets, without sacrificing the efficiency of hash learning, which is a necessary and favorable tradeoff. As discussed in the distribution analysis, remote sensing data suffer from dense-sparse heterogeneity, and simple linear projections fail to capture the complex nonlinear alignments, resulting in mediocre retrieval accuracy. Simpler baselines achieve lower mAP due to inadequate cross-modal interaction, while **CTHRS** maintains efficiency in retrieval and is at the same time level as other baselines.

In addition, different from large-scale vision Transformers that process high-resolution pixel patches, the proposed MSFT module operates exclusively on compact, preextracted feature vectors. Consequently, the additional matrix calculation introduced by the Transformer layers during inference is computationally negligible compared to the high computational cost of the feature extraction itself.

TABLE V
MAP RESULTS OF **CTHRS** AND BASELINES ON EXTREMELY SHORT CODE LENGTH

Task	Method	RSICD			UCM		
		5 bits	6 bits	7 bits	5 bits	6 bits	7 bits
I → T	DUCH	0.263	0.305	0.384	0.466	0.492	0.569
	CHNR	0.205	0.269	0.310	0.337	0.364	0.417
	JDSH	0.151	0.259	0.386	0.365	0.395	0.490
	DGCPN	0.201	0.253	0.288	0.197	0.250	0.269
	CTHRS	**0.423**	**0.505**	**0.546**	**0.624**	**0.668**	**0.700**
T → I	DUCH	0.271	0.331	0.383	0.453	0.506	0.545
	CHNR	0.196	0.255	0.305	0.290	0.330	0.369
	JDSH	0.168	0.250	0.370	0.394	0.445	0.555
	DGCPN	0.202	0.237	0.265	0.167	0.251	0.268
	CTHRS	**0.454**	**0.508**	**0.579**	**0.577**	**0.638**	**0.716**

F. Extremely Short Hash Code Length

In the perfect condition, the hash codes of instances that belong to the same category should be the same. Consequently, the total length of hash codes is $\log_2(c)$, where c is the number of tags. By forcing the model to operate with a minimal number of bits, we can assess the fundamental ability to preserve the critical semantic structure. Although short codes lead to insufficient capture of intraclass variations, remote sensing tasks often focus on coarse-grained classification, such as land use types, where short codes can balance accuracy and efficiency. However, most hashing methods are unable to guarantee a satisfactory retrieval performance with extremely short hash codes. In the experiments, code lengths are set to $\{5, 6, 7\}$ on RSICD and UCM datasets. As exhibited in Table V, the proposed **CTHRS** consistently outperforms the baselines, which is more suitable for large-scale remote sensing scenes.

From another perspective, short codes offer distinct advantages. Remote sensing data typically involve massive high-resolution images, and retrieval tasks require the processing of near-real-time data. The extremely short hash codes utilize Hamming distance for similarity calculation, and the query time is proportional to the code length. In reality, short codes can maintain a certain degree of distinguishability through optimization of contrast learning and neighborhood structure preservation in **CTHRS** while accelerating the retrieval process. In the cross-modal remote sensing scene, which contributes to quickly associating spatiotemporal heterogeneous data and achieving efficient knowledge discovery.

G. Distribution and Embedding Visualization

The complete period of feature distributions on RSITMD and UCM datasets is depicted in Fig. 7. Specifically, for original remote sensing images and text features, there exist significant modality-specific differences. After modal semantic fusion based on self-attention and cross-modal hashing contrast, the feature distributions become consistent in succession between heterogeneous representations.

In addition, the t-distributed stochastic neighbor embedding (t-SNE) is utilized to visualize diverse embeddings on the RSICD dataset, as demonstrated in Fig. 8. Compared with original embeddings, obvious separation is achieved after

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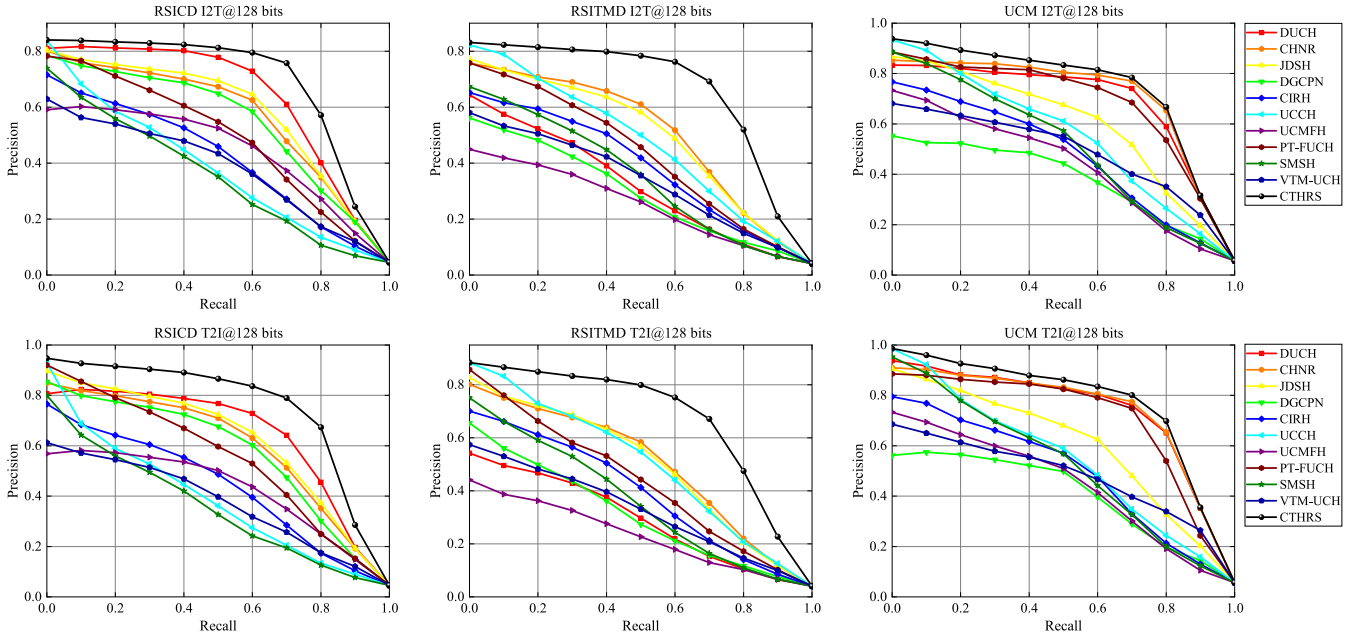


Fig. 5. PR curves of **CTHRS** and baselines on RSICD, RSITMD, and UCM datasets with a code length of 128 bits.

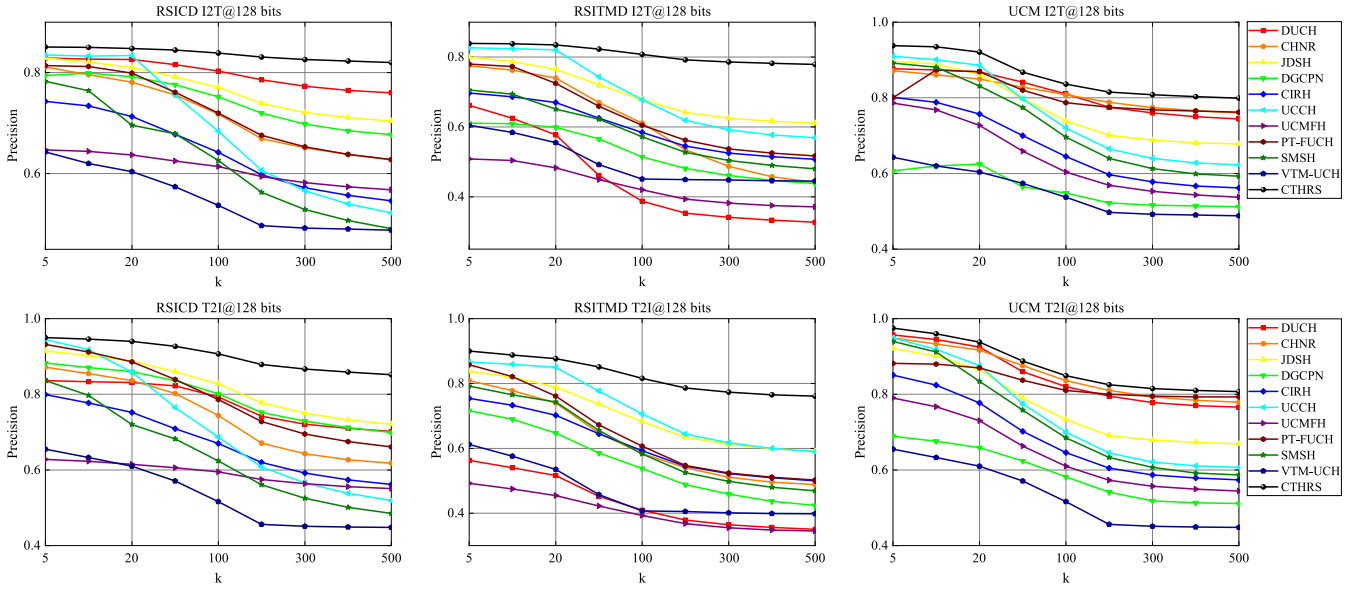


Fig. 6. Top-k curves of **CTHRS** and baselines on RSICD, RSITMD, and UCM datasets with a code length of 128 bits.

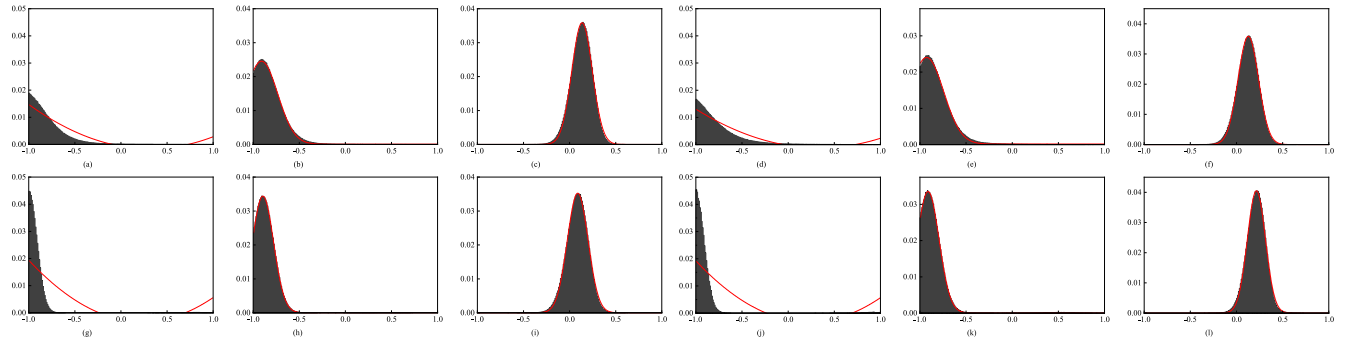


Fig. 7. Histogram distribution of $\mathbf{S}_I$ and $\mathbf{S}_T$ on RSITMD and UCM datasets. (a) S_I and (g) S_T on RSITMD. (b) $S_{I\text{Transformer}}$ and (h) $S_{T\text{Transformer}}$ on RSITMD. (c) H_I and (i) H_T on RSITMD. (d) S_I and (j) S_T on UCM. (e) $S_{I\text{Transformer}}$ and (k) $S_{T\text{Transformer}}$ on UCM. (f) H_I and (l) H_T on UCM.

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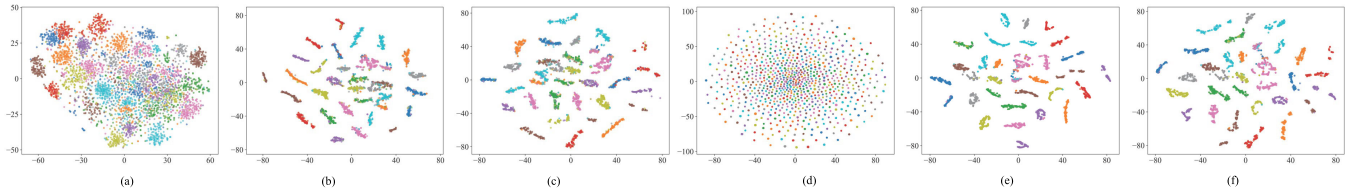


Fig. 8. t-SNE visualization on the RSICD dataset with a code length of 128 bits. (a) Original image embeddings. (b) Image embeddings of semantic fusion. (c) Image embeddings of hashing contrast. (d) Original text embeddings. (e) Text embeddings of semantic fusion. (f) Text embeddings of hashing contrast.

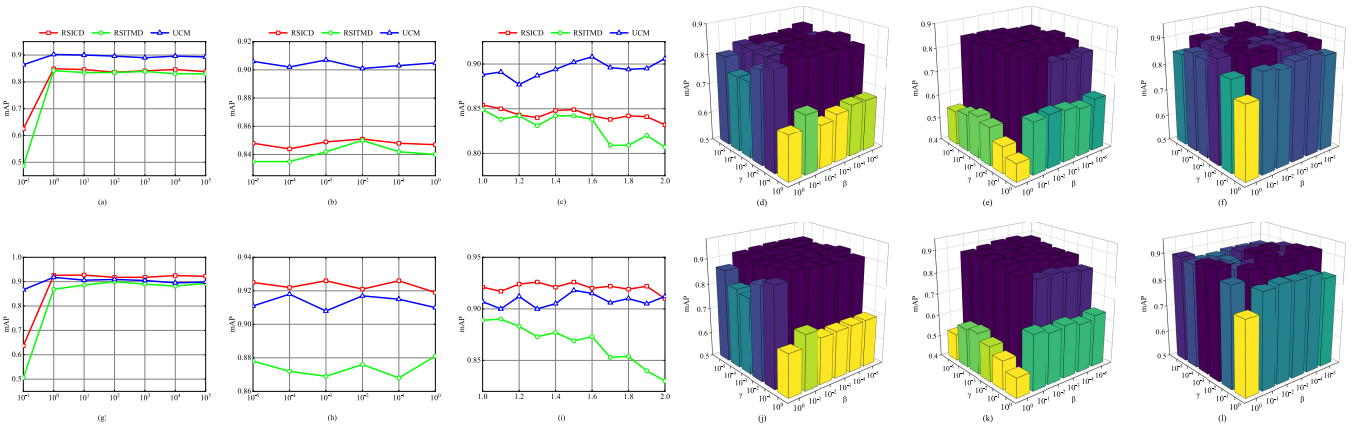


Fig. 9. Hyperparameter analysis of $\alpha, \beta, \lambda, \gamma$, and η on RSICD, RSITMD, and UCM datasets. (a) α @ I2T. (b) λ @ I2T. (c) η @ I2T. (d) RSICD @ I2T. (e) RSITMD @ I2T. (f) UCM @ I2T. (g) α @ T2I. (h) λ @ T2I. (i) η @ T2I. (j) RSICD @ T2I. (k) RSITMD @ T2I. (l) UCM @ T2I.



Fig. 10. Top-20 retrieval results on the RSICD dataset.

performing the proposed MSFT and cross-modal hashing contrast, that is, the intermediate and final embeddings, which demonstrates that **CTHRS** exhibits superior discriminability in a common latent space.

H. Parameters' Sensitivity Analysis

To verify the impacts of hyperparameters on diverse datasets, a comprehensive sensitivity analysis of $\alpha, \beta, \lambda, \gamma$, and η is presented. Particularly, the values of $\alpha, \beta, \lambda, \gamma$, and η are assigned in grid search, while the other parameters are fixed, and the hash code length is set to 64 bits. The mAP results are depicted in Fig. 9; it can be found that **CTHRS** obtains stable and superior retrieval accuracy under a range of parameters on the three datasets simultaneously, that is, $\alpha \in [10^0, 10^5]$, $\beta \in [10^{-5}, 10^{-1}]$, $\lambda \in [10^{-5}, 10^0]$, $\gamma \in [10^{-5}, 10^{-1}]$, and $\eta \in [1.2, 1.6]$. Therefore, the parameters are insensitive to cross-modal tasks over a broad range of values, which exhibits the generalization of **CTHRS**.

I. Case Study

This section exhibits the practical case of the top-20 image retrievals from image and text queries via **CTHRS** with a code length of 128 bits on the RSICD dataset. Instances with red marks are retrieval errors, which are distinctive from the others. Particularly, the query text semantically contains countryside scenery such that farmland and conspicuous lines are involved, and the correct samples contain the corresponding elements.

The MSFT module relies on global representations that lack fine-grained alignment. For visually uniform textures, it

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captures dominant background semantics but loses localized details. In the I2T case, the global fusion vector compresses a heavily textured "Farmland" image into generic "vegetation," and failing to distinguish crops from trees, the model incorrectly retrieves two descriptions mentioning "forest." To bridge the distribution gap, the IDCL module uses KL divergence to expand the sparse text toward dense visual targets. However, when a query contains rigid geometric constraints, forcing it to approximate a broader visual topology causes semantic overstretching, where the text matches structurally analogous but semantically unrelated classes.

In consequence, Fig. 10 demonstrates that **CTHRS** performs superior in remote sensing scenes, overcoming the challenge of aligning heterogeneous information, such as vegetation and rustic areas with text descriptions that include geospatial attributes and specific use cases.

V. Conclusion

In this article, we propose a novel unsupervised hashing approach for cross-modal remote sensing retrieval, which integrates the Transformer architecture with contrastive learning to bridge the modality disparities and preserve semantic consistency in binary codes. To improve the discriminative competency of the model and generate compact hash codes, a comprehensive loss function that combines IDCL, CHCL, and similarity matrix regularization is proposed. Extensive experiments on three benchmark datasets demonstrate the superior performance of **CTHRS** to state-of-the-art hashing approaches. In future work, since hash learning plays a crucial part in cross-modal retrieval, it would be interesting to explore the dynamic feature fusion strategy based on self-attention with KL divergence for large-scale multisource modalities, e.g., SAR and monochromatic remote sensing images, and further analyze the semantic priority.

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